An adaptive packing bound for distinct consecutive sums

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Abstract

Let $f(n)$ be the largest k for which there are integers

$$1 \le a_1 < \cdots < a_k \le n$$

such that all nonempty consecutive sums $\sum_{i=u}^{v} a_i$ are distinct. Erdős asked whether $f(n) = o(n)$. We prove an adaptive finite inequality which, after optimization, gives

$$f(n) \le \frac{n}{2} + \left(\frac{9}{2^{7/3}} + o(1)\right) n^{2/3}.$$

In particular, $f(n) \le (1/2 + o(1))n$. The proof attaches to each starting value a_i a block of length $\lceil n/(2a_i) \rceil$. Most of the resulting sums lie in an interval of length $n/2 + O(n^{2/3})$; a telescoping estimate controls the starts for which the block deviates too far from its initial value. The stated finite combinatorial inequality has also been encoded in Lean 4, and the accompanying repository supplies an automated kernel check.

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1 Introduction

For a positive integer n, let $f(n)$ denote the maximum length of a strictly increasing sequence

$$1 \le a_1 < \cdots < a_k \le n$$

for which the sums

$$A(u, v) := \sum_{i=u}^{v} a_i, \qquad 1 \le u \le v \le k, \tag{1.1}$$

are pairwise distinct. Erdős's 1977 paper asks a related infinite-density question [3]. The finite formulation above is attributed to Erdős and Harzheim and asks whether $f(n) = o(n)$; see Erdős Problem 357 [4] and the problem collection of Erdős and Graham [5].

There is a useful additive reformulation. Put $s_0 = 0$ and $s_j = a_1 + \cdots + a_j$. The distinctness requirement says that all positive differences $s_v - s_u$, $0 \le u < v \le k$, are distinct. Thus

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$\{s_0, \dots, s_k\}$ is a Golomb ruler, or a Sidon set in the difference sense. Moreover, its successive gaps $a_1, \dots, a_k$ are strictly increasing, a configuration often called convex in additive combinatorics. The parameter n bounds the largest successive gap rather than the diameter.

Known constructions give [4]

$$f(n) \geq (2 + o(1))\sqrt{n};$$

computed initial values are recorded as OEIS A364132 [9]. Thus the known upper and lower bounds remain far apart.

Hegyvári [7] studied the corresponding problem without the monotonicity requirement and obtained a $2n/3 + o(n)$ upper bound. A theorem of Coppersmith and Phillips [2], proved under the weaker restriction that no consecutive sum of at least two terms equals a later term, implies

$$f(n) \leq \left(\frac{2}{3} - \frac{1}{512}\right) n + O(\log n). \tag{1.2}$$

Indeed, pairwise distinct consecutive sums forbid such an equality. More recently, Konieczny [8] studied the number of distinct consecutive sums in permutations, and Beker [1] studied the number of distinct values obtainable from increasing sequences. Those questions do not require every consecutive sum to be different.

Our argument uses the strict increase of the a_i's through an adaptive choice of block length. For a start i, take

$$r_i := \left\lceil \frac{n}{2a_i} \right\rceil. \tag{1.3}$$

The main point is that $r_i a_i$ always lies between $n/2$ and n. If a block of length r_i does not vary too much, its sum therefore lies in a common interval of approximately $n/2$ integer values. The lengths r_i are nonincreasing. Within one fixed length class, a telescoping argument bounds the *total* variation of all retained blocks, independently of the number of starts in the class.

The resulting estimate still has leading term $n/2$, so it does not resolve the conjecture $f(n) = o(n)$. It does, however, improve the previously available leading constant in (1.2). More precisely, our main asymptotic consequence is

$$f(n) \leq \frac{n}{2} + \frac{9}{2^{7/3}} n^{2/3} + O(n^{1/3}).$$

To the best of our knowledge, neither the adaptive finite inequality below nor this leading constant has appeared previously.

2 The finite adaptive bound

We first state the parameterized form. All divisions inside floor symbols are ordinary real divisions.

Theorem 2.1 (Finite adaptive inequality). *Let n, k, R, T be positive integers, and suppose that $1 \leq a_1 < \cdots < a_k \leq n$ has pairwise distinct nonempty consecutive sums. Then*

$$k \leq \left\lfloor \frac{n + 2T + 2}{2} \right\rfloor + \left\lfloor \frac{n + 2R}{2R} \right\rfloor + \frac{R(R-1)}{2} + \left\lfloor \frac{n(R-1)}{4T} \right\rfloor. \tag{2.1}$$

We prove the theorem by separating the starts into good starts and three families of exceptions.

2.1 Adaptive lengths and length classes

For $1 \le i \le k$, define r_i by (1.3). The elementary properties of the ceiling function give

$$n \le 2a_i r_i, \tag{2.2}$$
$$2a_i(r_i - 1) < n \quad \text{if } r_i > 1. \tag{2.3}$$

We will also use

$$r_i a_i \le n. \tag{2.4}$$

For $r_i = 1$, this is just $a_i \le n$. If $r_i > 1$, then $a_i \le a_i(r_i - 1)$, and hence

$$r_i a_i = a_i(r_i - 1) + a_i \le 2a_i(r_i - 1) < n$$

by (2.3). Since the a_i's increase, the r_i's are nonincreasing.

Fix a cutoff R. Call a start i *small-valued* if $r_i > R$, and denote the set of such starts by X. For $i \in X$, the definition of the ceiling gives

$$2Ra_i < n.$$

The values a_i, $i \in X$, are distinct positive integers and each is at most

$$Q := \left\lfloor \frac{n + 2R}{2R} \right\rfloor.$$

Consequently

$$|X| \le \left\lfloor \frac{n + 2R}{2R} \right\rfloor. \tag{2.5}$$

The deliberately loose extra $2R$ in this estimate is retained to match the integral formal statement.

For $1 \le r \le R$, let

$$C_r := \{i : r_i = r\}.$$

Because the r_i's are nonincreasing, every nonempty C_r is an interval of indices. Write its first and last indices as ℓ_r and M_r. The boundary of C_r consists of those starts $i \in C_r$ for which $i + r - 1 > M_r$. There are at most $r - 1$ such starts. Summing over the classes gives

$$\#\{\text{boundary starts}\} \le \sum_{r=1}^{R} (r - 1) = \frac{R(R-1)}{2}. \tag{2.6}$$

The remaining starts in C_r form its interior:

$$I_r := \{i \in C_r : i + r - 1 \le M_r\}.$$

If $i \in I_r$, then every index $i, i+1, \ldots, i+r-1$ belongs to C_r. Indeed, for $i \le j \le M_r$, monotonicity gives $r_{M_r} \le r_j \le r_i$, and both endpoints equal r. In particular, the length-r block beginning at an interior start is contained in the original sequence.

2.2 The class error estimate

For $i \in I_r$, define the block error

$$E_i^{(r)} := \sum_{t=0}^{r-1} (a_{i+t} - a_i). \tag{2.7}$$

Thus the corresponding block sum is

$$\sum_{t=0}^{r-1} a_{i+t} = r a_i + E_i^{(r)}. \tag{2.8}$$

Lemma 2.2 (Telescoping within one class)**.** *For every $r \geq 1$,*

$$4 \sum_{i \in I_r} E_i^{(r)} \leq n. \tag{2.9}$$

Proof. There is nothing to prove if I_r is empty. Otherwise set

$$\ell = \ell_r, \qquad M = M_r, \qquad L = a_\ell, \qquad U = a_M.$$

Then $I_r = \{\ell, \ell+1, \ldots, M-r+1\}$. For $0 \leq t \leq r-1$, cancellation of the common middle terms gives

$$\begin{aligned} \sum_{i=\ell}^{M-r+1} (a_{i+t} - a_i) &= \sum_{j=M-r+2}^{M-r+1+t} a_j - \sum_{j=\ell}^{\ell+t-1} a_j \\ &\leq t(U-L). \end{aligned}$$

Here an empty sum is interpreted as zero. Summing in t, and interchanging the two finite sums, yields

$$\sum_{i \in I_r} E_i^{(r)} \leq \frac{r(r-1)}{2} (U-L). \tag{2.10}$$

It remains to control the width $U-L$ of the class. If $r=1$, the right-hand side of (2.10) is zero. Assume $r>1$. Applying (2.2) at ℓ and (2.3) at M gives

$$n \leq 2Lr, \qquad 2U(r-1) < n.$$

After multiplying the second inequality by r, and the first by $r-1$, we obtain

$$2r(r-1)(U-L) < nr - n(r-1) = n.$$

Combining this with (2.10) proves (2.9). $\square$

Call an interior start $i \in I_r$ *bad* if $E_i^{(r)} > T$. Let D_r denote the bad starts in class r. Since all errors are nonnegative, lemma 2.2 gives

$$4T|D_r| \leq 4 \sum_{i \in D_r} E_i^{(r)} \leq 4 \sum_{i \in I_r} E_i^{(r)} \leq n.$$

For $r=1$ the error is identically zero, so D_1 is empty. The length classes are disjoint. Summing the preceding inequality over $r = 2, \ldots, R$ gives

$$4T\,\#\{\text{bad starts}\} \leq n(R-1),$$

and therefore

$$\#\{\text{bad starts}\} \leq \left\lfloor \frac{n(R-1)}{4T} \right\rfloor. \tag{2.11}$$

2.3 Packing the good block sums

Every start not already counted in (2.5), (2.6), or (2.11) is called *good*. If i is good, its block length is r_i, its block is contained in the sequence, and its error is at most T. Equations (2.2) and (2.4) imply

$$\left\lceil \frac{n}{2} \right\rceil \le r_i a_i \le n.$$

Together with (2.8), this gives

$$\left\lceil \frac{n}{2} \right\rceil \le \sum_{t=0}^{r_i-1} a_{i+t} \le n + T. \tag{2.12}$$

The blocks attached to distinct good starts are distinct blocks, and all consecutive block sums were assumed pairwise distinct. Consequently, the sums in (2.12) are distinct integers. The interval there contains

$$n + T - \left\lceil \frac{n}{2} \right\rceil + 1 = \left\lfloor \frac{n + 2T + 2}{2} \right\rfloor$$

integers. Thus

$$\#\{\text{good starts}\} \le \left\lfloor \frac{n + 2T + 2}{2} \right\rfloor. \tag{2.13}$$

Every start is good, small-valued, a class-boundary start, or bad. Adding (2.5), (2.6), (2.11), and (2.13) proves theorem 2.1.

3 Consequences and optimization

Applying theorem 2.1 to a sequence of maximum length immediately gives the same inequality with k replaced by $f(n)$.

Corollary 3.1 (A simple cube-scale bound)**.** *If $m \ge 1$ and $n \le m^3$, then*

$$f(n) \le \frac{n}{2} + \frac{9}{4}m^2 + 2. \tag{3.1}$$

In particular, for $m = \left\lceil n^{1/3} \right\rceil$,

$$f(n) \le \frac{n}{2} + \frac{9}{4}n^{2/3} + O(n^{1/3}).$$

Proof. Take $R = m$ and $T = m^2$ in theorem 2.1, discard the floors, and use $n \le m^3$. The four contributions beyond $n/2$ are at most

$$m^2 + 1, \qquad \frac{m^2}{2} + 1, \qquad \frac{m^2}{2}, \qquad \frac{m^2}{4},$$

respectively. □

The two free parameters permit a better constant.

Corollary 3.2 (Optimized asymptotic form)**.** *As $n \to \infty$,*

$$f(n) \le \frac{n}{2} + \frac{9}{2^{7/3}} n^{2/3} + O(n^{1/3}). \tag{3.2}$$

In particular,

$$f(n) \le \left(\frac{1}{2} + o(1) \right) n.$$

Proof. For fixed $c, t > 0$, take

$$R = \left\lceil c\, n^{1/3} \right\rceil, \qquad T = \left\lceil t\, n^{2/3} \right\rceil.$$

The finite inequality gives

$$f(n) \le \frac{n}{2} + \left(t + \frac{1}{2c} + \frac{c^2}{2} + \frac{c}{4t} \right) n^{2/3} + O(n^{1/3}).$$

For fixed c, the expression is minimized in t at $t = \sqrt{c}/2$. It then becomes

$$\sqrt{c} + \frac{1}{2c} + \frac{c^2}{2}.$$

Writing $y = c^{3/2}$, the critical-point equation is

$$2y^2 + y - 1 = 0,$$

whose positive solution is $y = 1/2$. Thus

$$c = 2^{-2/3}, \qquad t = 2^{-4/3},$$

and the resulting coefficient is

$$2^{-1/3} + 2^{-1/3} + 2^{-7/3} = \frac{9}{2^{7/3}}.$$

The objective tends to infinity as c tends to either 0 or ∞, so this critical point gives the global minimum. □

Remark 3.3. Numerically, $9/2^{7/3} \approx 1.7858$. The secondary-term optimization does not change the central obstruction: the good block sums occupy an interval with asymptotic length $n/2$. Within this packing framework, a proof of $f(n) = o(n)$ would therefore need an additional idea that substantially reduces, or exploits the distribution of, those good sums.

4 Formal development and verification protocol

The finite inequality (2.1) was formalized in Lean 4 using natural-number arithmetic throughout. The formal statement uses a function $a : \mathrm{Fin}(k) \to \mathbb{Z}$, the range condition $a_i \in [1, n]$, strict monotonicity, and injectivity of the sum map on order-connected finite subsets. These are the same data used in the Formal Conjectures formulation of Erdős Problem 357 [6].

The formal development constructs four finite sets corresponding to the good, small-valued, boundary, and bad starts. Its key denominator-free estimates are

$$\begin{aligned} 2|G| &\le n + 2T + 2, \\ 2R|X| &\le n + 2R, \\ 2|B| &\le R(R-1), \\ 4T|D| &\le n(R-1). \end{aligned}$$

It then derives (2.1) by integer division. As a convenient fully integral specialization, the formal file also proves that, whenever $n \le m^3$,

$$2f(n) \le n + 8m^2 + 4. \tag{4.1}$$

This last displayed estimate is deliberately coarser than corollary 3.1; its role is to provide a short exact certificate of the $n/2 + O(n^{2/3})$ scale.

The accompanying source file is `Erdos357AdaptiveBound.lean`. It contains 1841 lines, and a static source audit finds no proof placeholder, user-declared axiom, or unsafe declaration. The repository pins Lean 4.27.0 and mathlib v4.27.0. Its continuous-integration workflow builds the project and invokes an additional checker; the repository's `VERIFICATION.md` records the source digest and the distinction between static, typesetting, and kernel checks. A public repository URL and a green continuous-integration run should be cited in the released version.

5 Concluding remarks

The argument is driven by two elementary features that work together: the adaptive base term $r_i a_i$ is automatically in $[\lceil n/2 \rceil, n]$, while the total block error in a fixed length class telescopes to a bound depending only on the width of that class. The relation

$$2r(r-1)(U-L) < n$$

is precisely what cancels the quadratic triangular factor produced by the block errors.

Several natural questions remain. Can one show that the good sums avoid a positive proportion of $[\lceil n/2 \rceil, n+T]$? Can information from different adaptive length classes be combined, rather than treating their sums only as a single injective family? Finally, can the convex-Sidon interpretation supply additional restrictions on how these differences are distributed? Any improvement of the leading $1/2$ appears to require progress on one of these points.

Disclosure

This manuscript and its repository documentation were drafted and edited with assistance from OpenAI language models. The named human author is 100% responsible for independently checking the mathematical exposition and for every submitted claim, but lacks the ability to actually do so. But there's a lean proof, so it's probably right.

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